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WorldFlow

Joint Video and 3D Motion Prediction via Large-Scale Pretraining

Video + query-point 3D world model

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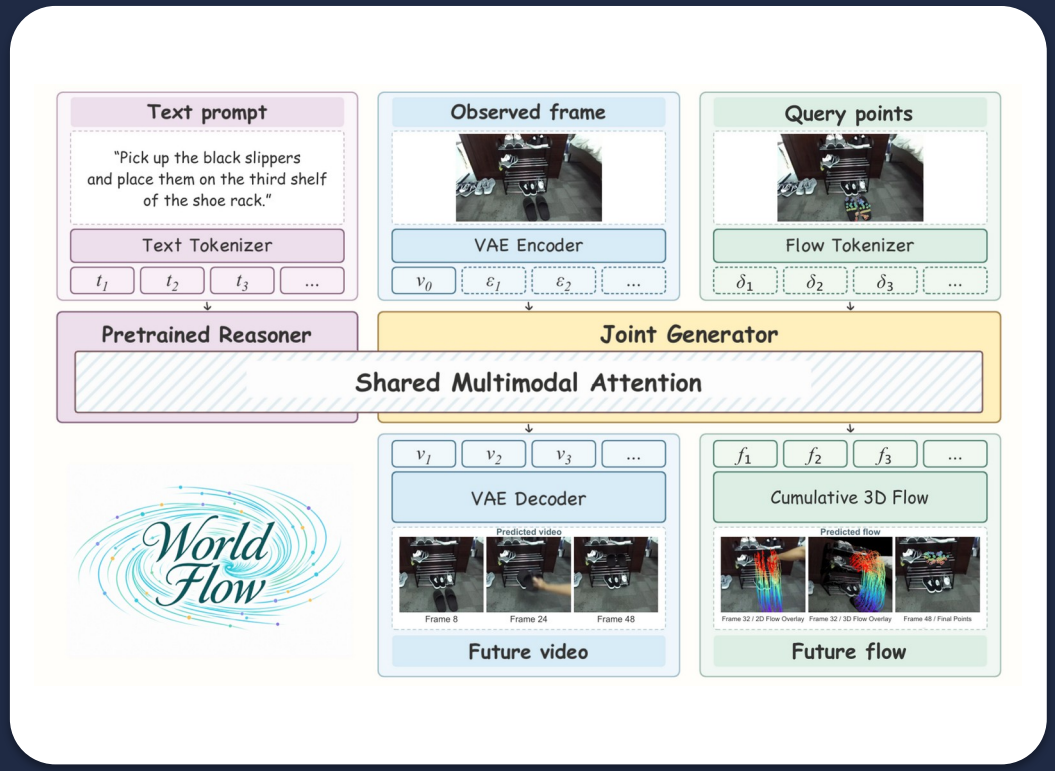


Image: paper Fig. 2 (WorldFlow overview)

One-line summary

Adds a “3D displacement token” modality to the pretrained video model Cosmos-Nano, using one MoT generator to jointly denoise future video and metric 3D trajectories of query points; joint video generation nearly halves 3D motion error, yet on the real-world subset it still trails the “zero-motion” reference.

Each prior line lacks explicit 3D, cross-embodiment, or supervision

Video world models

Wan • Seedance • Cosmos

Strengths

Predict increasingly coherent future frames

Gaps

Pixels only; 3D dynamics stay implicit, hard to use directly for planning and control

World-act. WAM

Predict observations and actions jointly

Strengths

Brings actions into world modeling for end-to-end control

Gaps

Action labels are costly and tied to specific robot embodiments

3D motion prediction

MolmoMotion • μ0 • TraceGen

Strengths

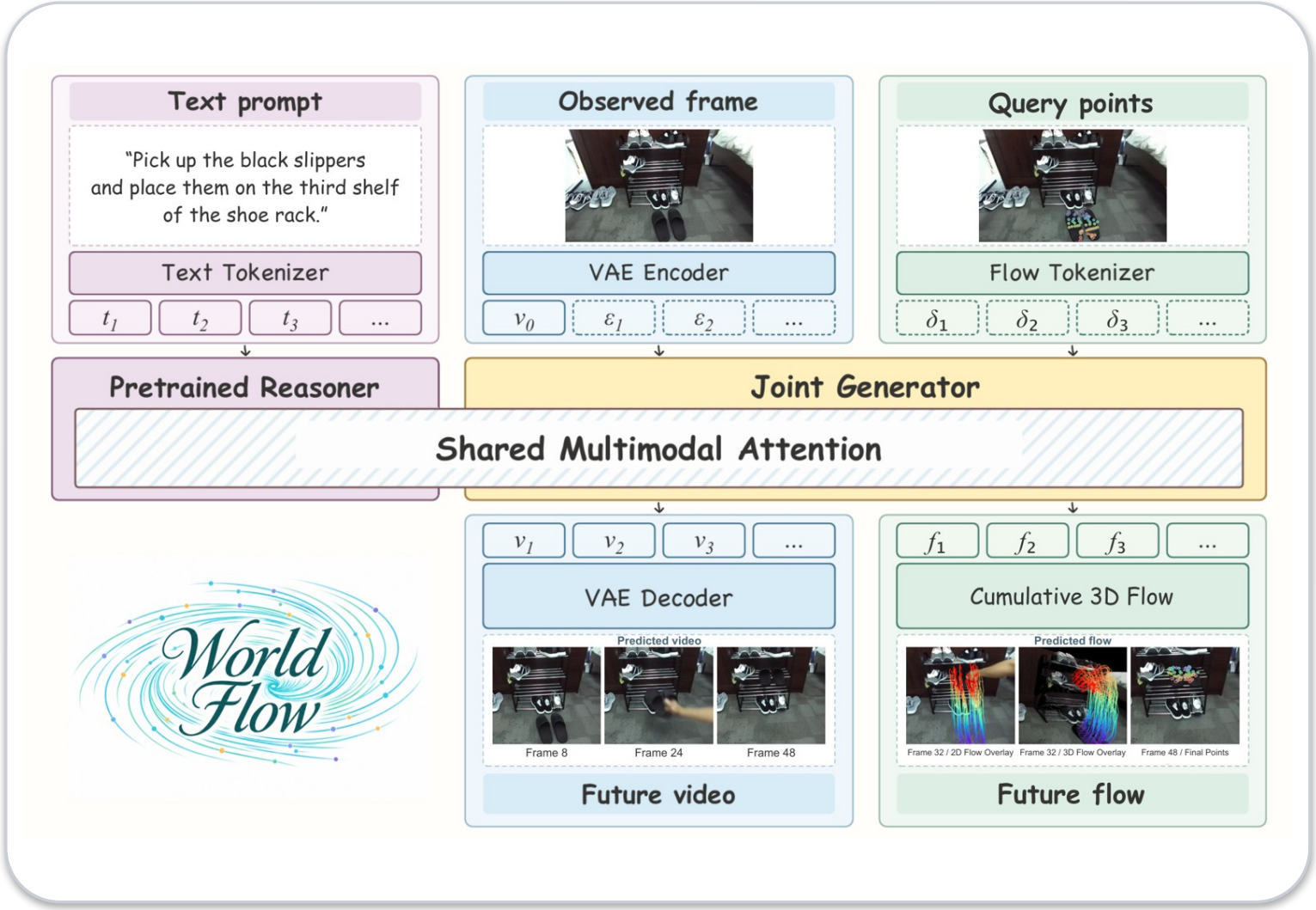
Directly predict future 3D trajectories/flow

Gaps

Flow-only supervision is sparse; dense scene flow is costly to compute

WorldFlow's idea: treat "query-conditioned 3D flow" as an embodiment-agnostic geometric interface and **generate it jointly** with video—dense video supervision feeds motion learning, directly exploiting the physical priors of pretrained video models.

Joint denoising of future video + 3D tracks, no tracker stage



1 Input

Instruction c, first frame I_0, N 3D query points and their pixel coordinates

2 Joint denoising

Video and flow tokens interact via full attention and read the Reasoner's language context; both modalities share a noise level

3 Output

VAE decodes future video; per-frame displacements f_t, de-normalized and summed → metric 3D tracks (first-frame camera frame)

Objective: $p(I_{1:t}, f_{1:t} \mid I_0, Q, S_0, c)$

3D flow tokens: continuous offsets + geometry + video ST RoPE

① Per-domain norm.

$$f = f / a_d$$

a_d: a per-domain scalar, pre-estimated and fixed during training, balancing motion scales across domains

② Query geometry

$$e^{3D} = \mathrm{MLP}[E_abs(p_0); E_rel(p_0 - \mu)]$$

Absolute and centroid-relative positions each pass through Fourier features + MLP; shared by all future frames of a point

③ Continuous tokens

$$h = \mathrm{Linear}(f^{(\sigma)}) + e^{3D} + e_cond(\sigma)$$

One continuous flow token per "point × frame"; e_cond includes modality and noise-level embeddings

④ ST-aligned RoPE

$$r = (t/q_t, \lfloor v_0/q_s p_s \rfloor, \lfloor u_0/q_s p_s \rfloor)$$

Anchored to the future frame and the point's first-frame patch, sharing the position rule of video tokens

13 × 9 × 15
Video token grid per clip

48 × 64
flow tokens (48-step displacements × 64 query points)

49 frames • 288×480
q_t=4, q_s=16, p_s=2

Objective: joint rectified flow + 3 auxiliary losses on clean motion

$$L = \lambda_{video} \cdot L_{video} + \lambda_{flow} \cdot L_{flow} + 0.1 \cdot L_{dir} + 0.1 \cdot L_{vmag} + 0.05 \cdot L_{end}$$

Joint flow matching: video and flow share a per-sample noise level, with separate losses; **Auxiliary terms:** applied to the reconstructed clean flow $\hat{x}_0 = x_\sigma - \sigma \cdot v_\theta$

L_vmag step displacement

$$\| \hat{f}_{ti} - f_{ti} \|^2$$

Constrains local displacement magnitude and direction; provably equivalent to σ^2-reweighted vector-field error

L_dir cumulative direction

$$1 - \cos(\hat{d}_{ti}, d_{ti})$$

Prevents small errors from accumulating into wrong directions; weighted by ground-truth cumulative displacement magnitude

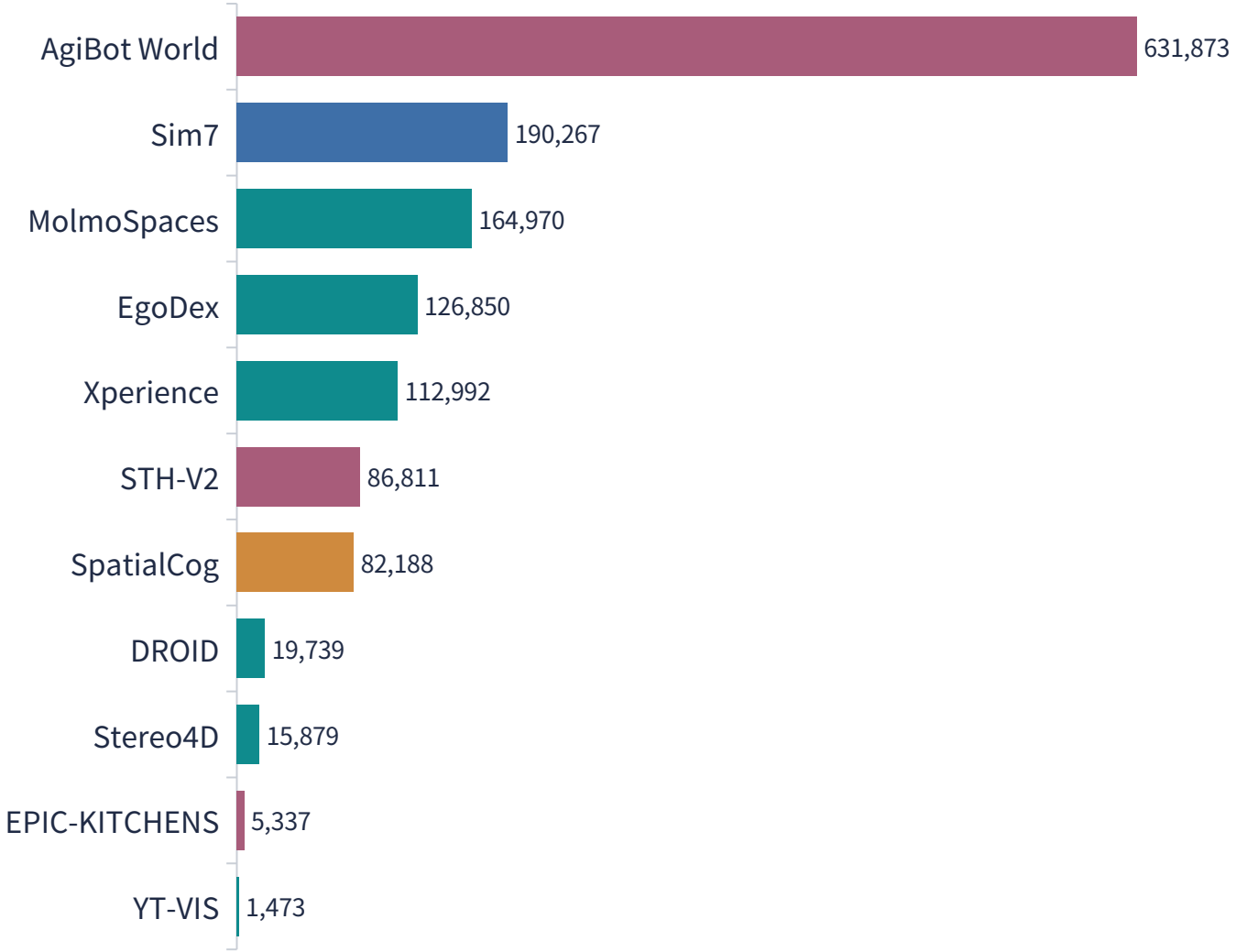
L_end relative endpoint

$$\| \hat{d}_T - d_T \| / \max(\| d_T \|, 0.1)$$

Constrains final position; points with net displacement < 0.1 (normalized coordinates) are excluded

All three auxiliary losses are weighted by 1−σ (more weight on low-noise samples); ablation: removing them raises Sim7 ADE 0.051 → 0.063 m

Training corpus: 1.44M object-level clips, ~44% from AgiBot World



Sim7 · Real3 · SpatialCog · MolmoMotion-1M

1.44M

Object-level clips

49 frames @16 FPS, 64 query points / clip

1,224 h

Std. duration

From clip count, not deduped raw video

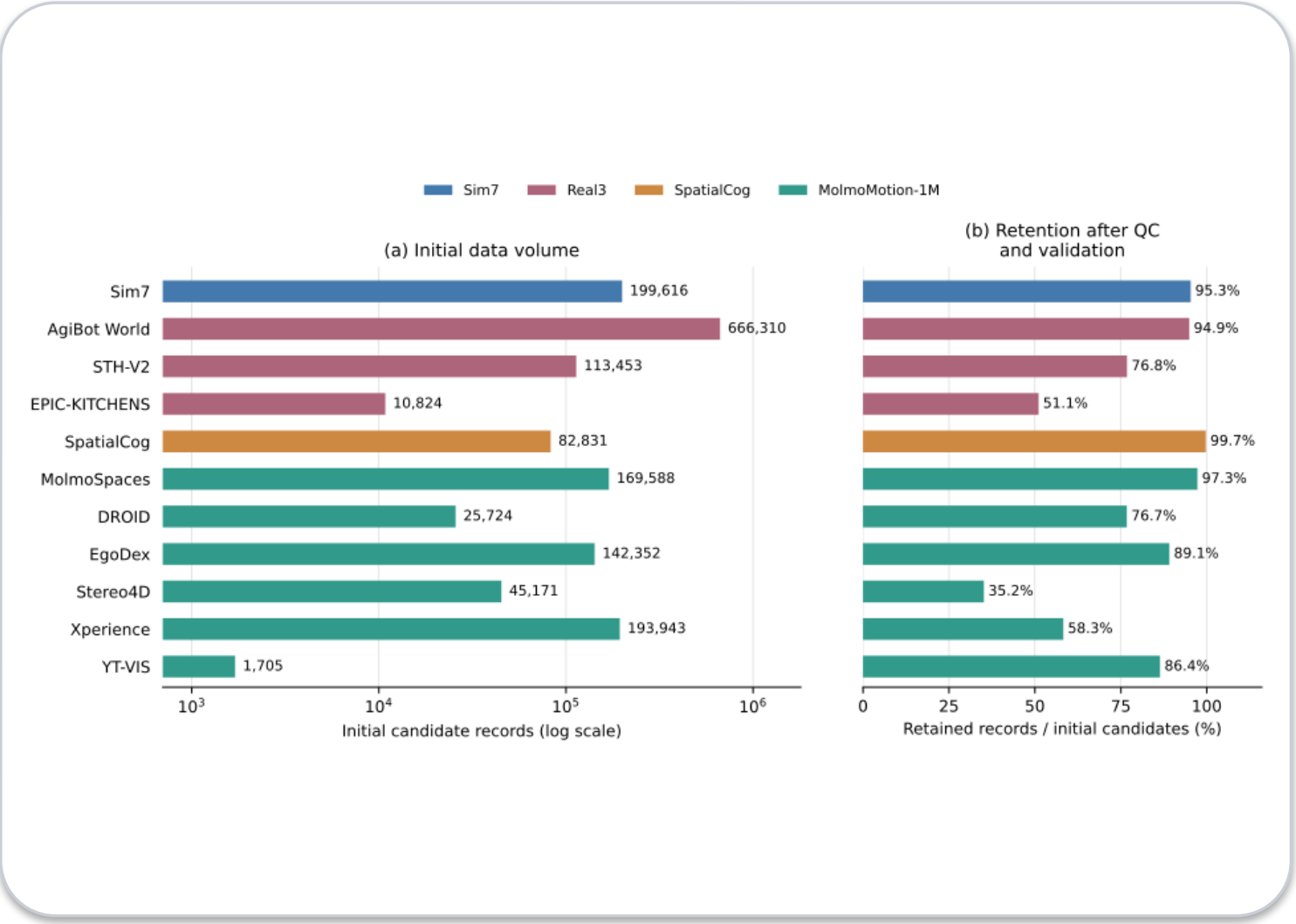
4.4B

3D displacement vectors

92.06M point tracks × 48 steps

Source: paper § 4.1, Appendix Table I

Diverse data: retention, length, motion scale per source



35.2%

Stereo4D has the lowest QC retention

SpatialCog 99.7%， AgiBot 94.9%， EPIC 51.1%（Fig. I）

97.8–99.7%

Three Molmo subsets need tail padding

MolmoSpaces / EgoDex / DROID; padded frames have zero displacement (Fig. II)

Several× to 100×

Cross-source motion-scale gap

Estimated from Fig. III; hence a fixed per-domain normalization coefficient a_d

2-stage: SpatialCog warm-up → sim → real → SpatialCog fine-tune

Stage 1

SpatialCog warm-up, 32k steps

Initialized from Cosmos-Nano; peak lr 1e-4, cosine decay

Stage 2

Mixed data, 51.1k steps

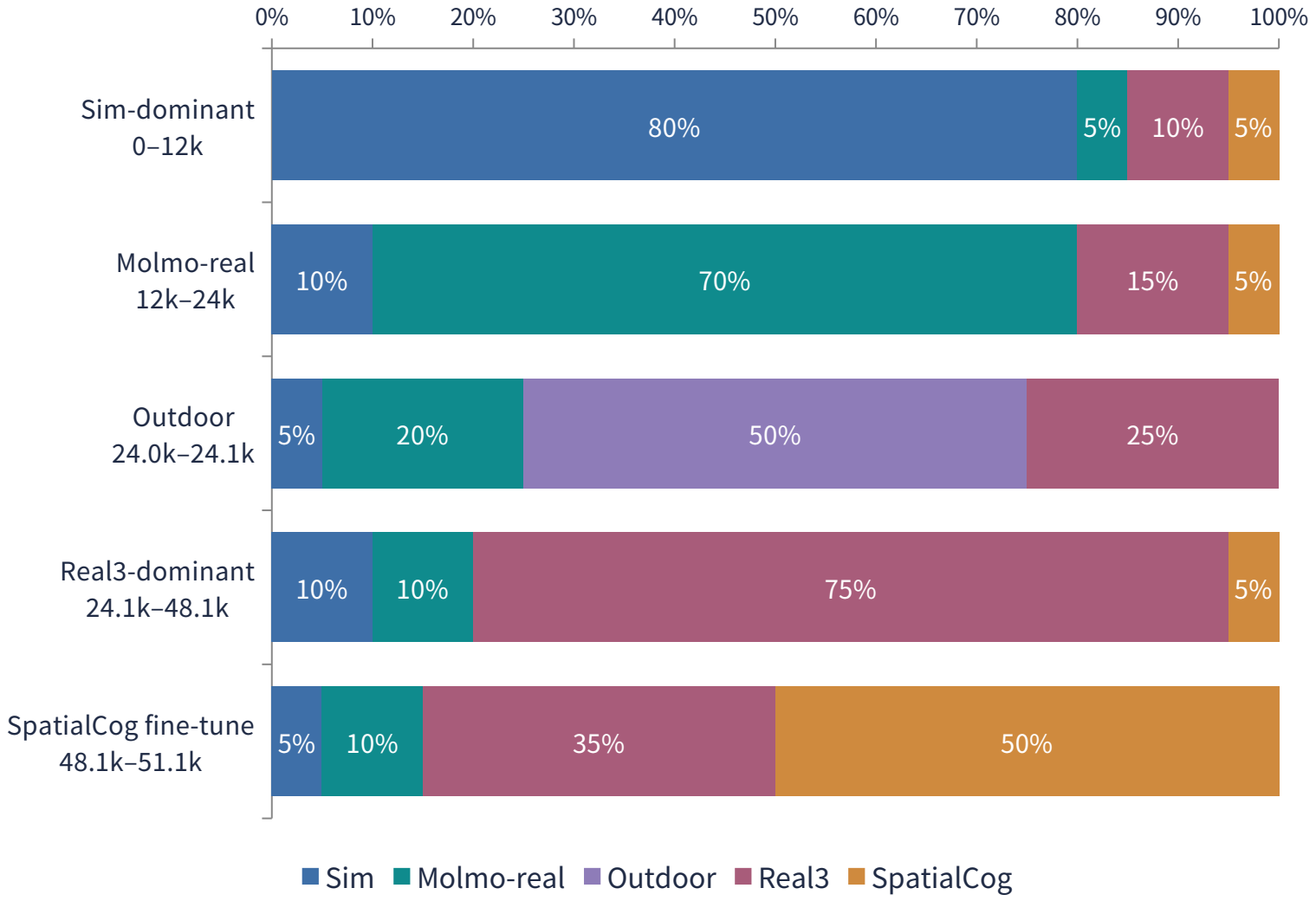
EMA-weight initialization; smooth transitions across five sampling phases

In-domain sampling p ∝ N^α (α=0.8 for Real3, 0.5 otherwise)

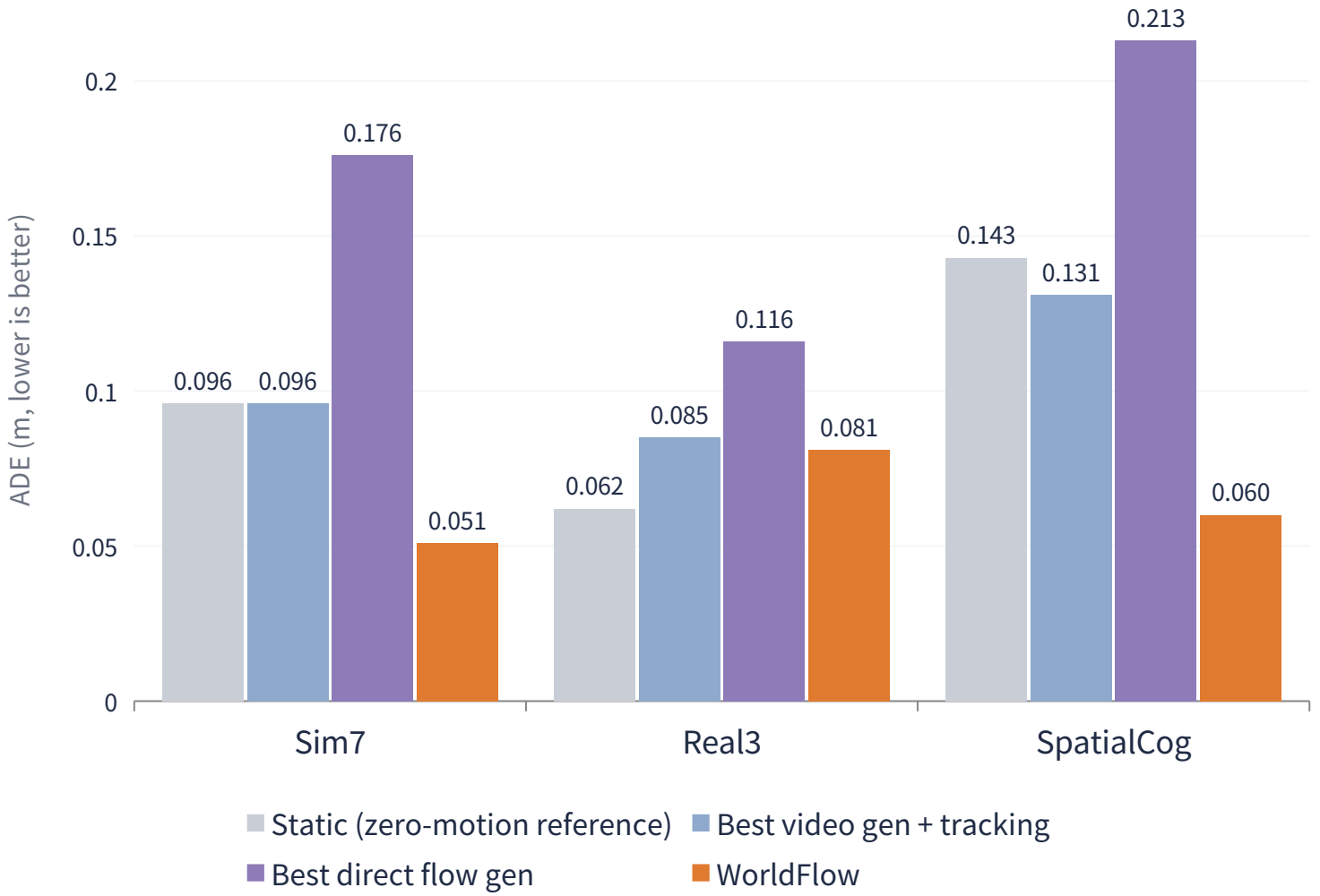
lr: 1e-4 → 8e-5 plateau → 5e-6 at the end

64 × H800, global batch 128

Inference: 40 sampling steps, CFG = 2



Main result: best learned ADE on all 3 subsets; SpatialCog −54%



ADE reduction vs. strongest external learned baseline

SpatialCog **−54.2%**

0.131 → 0.060 m (spatial-relation reasoning)

Sim7 **−46.9%**

0.096 → 0.051 m (7 simulation platforms)

Real3 **−4.7%**

0.085 → 0.081 m; yet Static is only 0.062

Strongest baselines: Cosmos3-Nano+4RC on Sim7 / Real3; LingBot-Video-MoE+4RC on SpatialCog

Source: paper Table 1 (WorldFlowBench; ADE in meters)

Ablation: joint video gen helps most; direct flow beats post-hoc tracking



Joint generation vs. flow only

−48.5% / −19.8% / −55.9%

ADE reduction (Sim7 / Real3 / SpatialCog); SpatialCog FDE 0.248 → 0.078 (−68.5%)

Direct flow vs. post-hoc tracking

0.060 vs 0.083 / 0.098

SpatialCog ADE: direct generation vs. T4W / 4RC tracking of the same video; better ADE, FDE, PWT on all three subsets

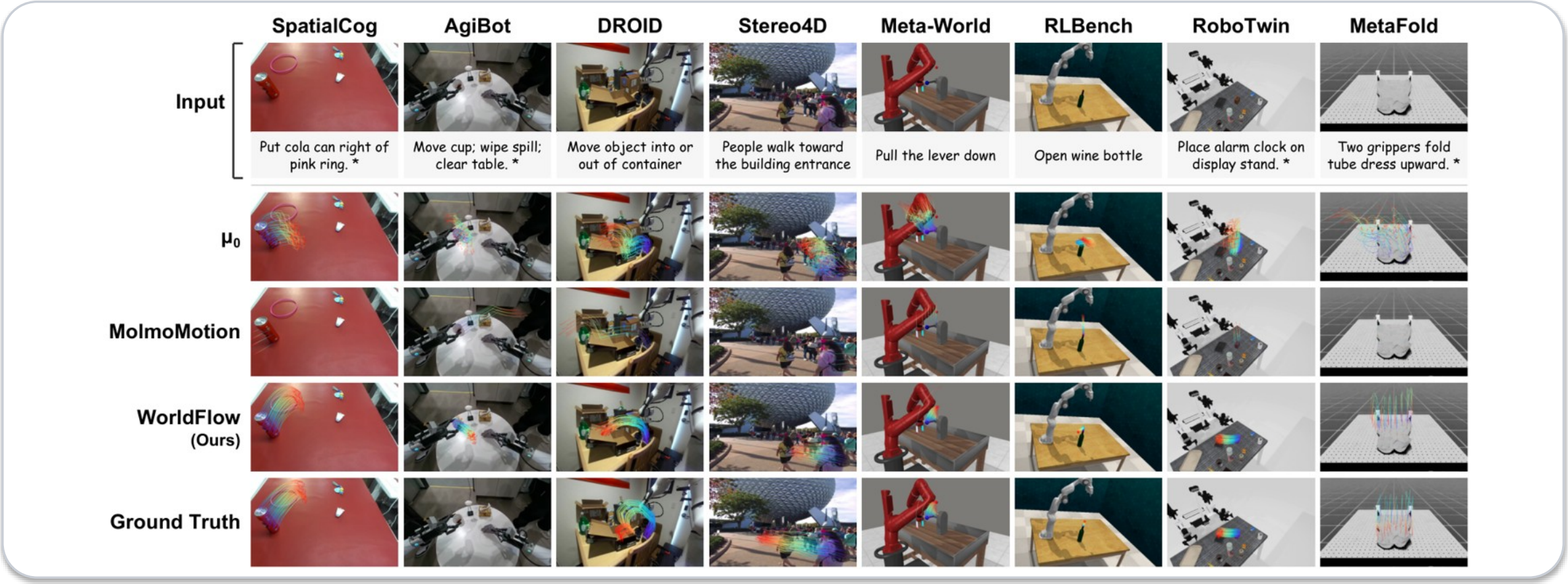
Auxiliary motion losses

Sim7 0.063 → 0.051

ADE −19%; ADE, FDE and PWT improve on all three subsets

Source: paper Table 1 ablations, § 4.3.3 (ADE in meters)

Qualitative: best GT shape match; open-scene motion underestimated



Shapes match ground truth

SpatialCog detour placement, Meta-World lever pulling, MetaFold skirt folding, etc.: trajectories closely match ground truth

Common baseline failures

μ0 trajectories scatter into clusters; MolmoMotion is often near-static or misdirected

Remaining weaknesses

Long-range pedestrian motion in Stereo4D is clearly underestimated; DROID arc amplitudes are biased

Source: paper Fig. 3 (trajectories projected onto input frames)

Real robot, flow predictor swapped: success 66.7% vs 11.1%

97.2%

Plausible flow predictions (35/36)

μ0： 25.0% （9/36）

66.7%

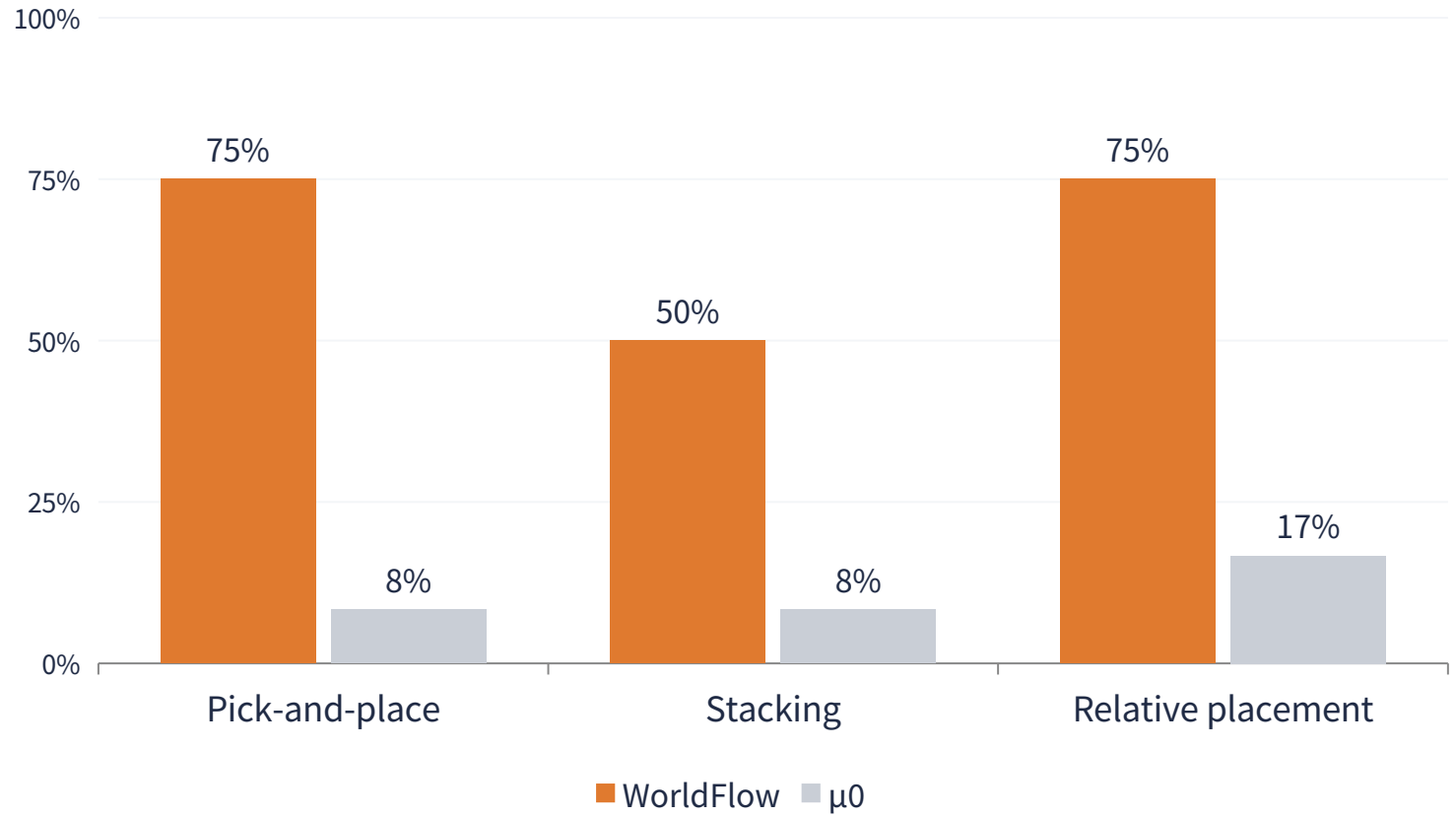
Real-robot success rate (24/36)

μ0： 11.1% （4/36）

Pipeline

Qwen localizes target → object point cloud as query points → WorldFlow predicts trajectories → T(R,O) Grasp grasping → local rigid fitting yields arm trajectory

Real-robot success rate by task group (4 tasks × 3 trials each)



For WorldFlow, “plausible prediction” and “execution success” differ by ~30 percentage points, mainly due to grasp failures, trajectory-optimization failures and unstable stacking placement.

Source: paper § 4.4, Table 2 (12 tasks × 3 trials = 36 trials / method)

Caution: on the real subset, “staying still” is more accurate

Static (zero motion) vs WorldFlow

Metric	Static	WorldFlow
Real3 ADE ↓	**0.062**	0.081
Real3 FDE ↓	**0.086**	0.113
Real3 PWT ↑	**0.464**	0.311
Sim7 PWT ↑	**0.573**	0.564
SpatialCog ADE ↓	0.143	**0.060**

Red: Static is better. On Real3, no learned method (incl. WorldFlow) beats the zero-motion reference; the paper excludes Static from rankings. On SpatialCog, WorldFlow clearly outperforms Static.

SpatialCog in-domain advantage

Self-collected data; WorldFlow is trained on it in the first and last stages, while baselines are zero-shot

Ground truth from automatic tracking

Sim7/Real3/SpatialCog tracks come from Track4World; training and evaluation share the source, so tracker bias may be learned

Heavy tail zero-padding

Many short clips padded to 49 frames, possibly biasing the model to underestimate late motion

Limited real-robot evidence

36 trials, compared only with μ0; “plausible flow” judged manually

Takeaway: explicit 3D motion as a video world model’s “2nd head”

1 Representation

Continuous “point × frame” displacement tokens + spatiotemporal RoPE shared with video plug into a pretrained video generator with minimal changes

2 Evidence

Controlled ablations are most convincing: joint video generation cuts ADE by 19.8–55.9%; direct flow generation beats post-hoc tracking

3 Significance

Flow is an embodiment-agnostic motion interface, complementary to WAMs; real-robot success 66.7% vs 11.1%

Open questions

Query points must be given in advance (authors plan automatic selection) · only a single future is modeled · whether and how much objects move in open real-world scenes remains hard to predict (Real3 trails Static) · joint generation with actions to serve policy learning